Date: January 19, 2026

Analysis of Body Mass Index (BMI) and Influencing Factors

This project is designed within the context of a public health and data analytics study aimed at understanding Body Mass Index (BMI) and the key demographic and lifestyle indicators that influence it. The dataset provided represents health-related information collected from more than 12,000 individuals, enabling large-scale analytical and statistical exploration of obesity-related risk factors.

The dataset contains variables including age, gender, height, weight, daily calorie intake, physical activity level, smoking status, alcohol consumption, and family history of obesity. BMI values have been calculated using standard medical formulas and categorized into underweight, normal, overweight, and obese classes. This structured dataset reflects realistic health data and is suitable for both exploratory data analysis and predictive modeling.

You have to submit two academic deliverables. The first deliverable is a comprehensive written report that critically examines the concept of BMI, its clinical significance, and the various biological and lifestyle factors affecting it. The report should also discuss how data analytics and machine learning techniques can support health assessment and early risk identification.

The second deliverable is a Google Colab notebook that demonstrates practical data analysis skills. You must preprocess the dataset, perform exploratory analysis, and apply a Logistic Regression model to classify individuals based on obesity risk. The notebook should include model evaluation and interpretation of results, highlighting the most influential variables contributing to BMI outcomes.